Daivik Wadhwani

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Education

Indian Institute of Information Technology Sricity **August 2023 – June 2027**
Bachelor of Technology in Computer Science *CGPA: 7.76*

Experience

Research Intern, HCI Lab **Jan 2026 – Present**

- Co-authored a research paper on one-handed VR accessibility for players with motor disabilities, accepted for presentation at ACM ASSETS 2026, the ACM SIGACCESS Conference on Computers and Accessibility.
- Designed and ran a user study with 17 participants using NASA-TLX (Raw TLX: 42.0) and the System Usability Scale (65.8); identified a negative learning slope in 12 of 13 long-session participants as quantitative evidence of motor adaptation. Publication

Projects

SplitSmart GitHub Live | *MERN, Socket.io, Gemini Vision API, JWT* **2026**

Collaborative group expense splitter with item-level bill claiming, automated OCR, and debt settlement.

- Implemented MongoDB version-key **optimistic concurrency control**: conditional `__v` updates and up to three retries with 50–100 ms jittered backoff prevented conflicting item claims; integrated **Gemini 2.5-Flash** OCR through in-memory Multer and an $O(N \log N)$ greedy cash flow solver.
- Optimized compound MongoDB indexes across four models, reducing expense document scans from **6,021 to 15** and database time from **9 ms to 1 ms**; added JWT refresh and rate limiting to prevent token exhaustion.

PowerPulse GitHub | *AWS Lambda, DynamoDB, API Gateway, Cognito, Serverless* **2025**

Serverless energy analytics system with rule-based and AI-driven insights.

- Architected a fully serverless backend on AWS Lambda, API Gateway, Cognito, and DynamoDB, eliminating idle infrastructure cost and enabling automatic scaling without capacity planning.
- Devised a caching strategy that cut AI response time from 1800 ms to 200 ms (89% faster) and reduced API costs by 70%; integrated Hugging Face inference with fallback control.

Bookish GitHub Live | *MERN, Redux Toolkit, Stripe, Upstash Redis, Jest* **2026**

Full-stack book marketplace connecting buyers, sellers, and admins through role-based dashboards.

- Integrated an Upstash Redis cache-aside layer across 11 endpoints, achieving an 84.82% average latency reduction with warm hits under 30 ms, down from 120–230 ms for cold database access.
- Implemented a CI/CD pipeline with GitHub Actions running 48 Jest tests across 12 suites on every push; added B-tree indexing, field projections, and `.lean()` hydration bypass, reducing CPU overhead by 35–50%.

Technical Skills

Languages: C, C++, C#, Java, JavaScript, HTML, CSS
Frontend: React, Tailwind CSS, Redux Toolkit, TanStack Query
Backend: Node.js, Express.js, REST APIs, Socket.io, JWT, Multer
Cloud & Infra: AWS Lambda, API Gateway, Cognito, Cloudinary, Render, Vercel
Databases: MongoDB, MySQL, DynamoDB, Upstash Redis
Tools: Git, GitHub, Jest, GitHub Actions, Unity, OpenXR, XR Interaction Toolkit, Postman

Achievements

- Secured 3rd place at Global Game Jam 2025 by shipping a complete game under a 48-hour constraint.
- Rated 1580 on LeetCode and solved 250+ problems across LeetCode and Codeforces.
- XR Lead, IOTA Club, leading XR initiatives, mentoring members, and managing lab projects.